

MARIA VITTORIA GURRIERI

POSTDOC RESEARCHER 📍 BERLIN, 12043, GERMANY 📞 +39 3294416534

◦ DETAILS ◦

Berlin, 12043
Germany

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marivi.gurrieri@gmail.com

Nationality
Italian

◦ PROGRAMMING ◦

Python

LaTeX

Office Package

Julia

Matlab

C/C++

SQL

◦ LANGUAGE SKILLS ◦

Italian

English

German

◦ HOBBIES ◦

I am an enthusiastic lover of outdoor activities.

I am passionate about travel and discovering new perspectives and environments. I am always ready to plan and organize trips.



PROFILE

I am a quantitative scientist with many years of experience in **data analysis**, **numerical modeling**, and **complex problem-solving**. During my academic career, I developed and applied advanced simulation to **analyze**, **interpret**, and **validate** results.

My plus is a strong **analytical mindset** and **attention to detail**. I am trained in innovative problem-solving, with the ability to translate abstract challenges into actionable, data-driven solutions.



PROFESSIONAL ACTIVITIES

PostDoc researcher at Humboldt University, Berlin

January 2025 — Present

- Performed **analytical computations** in quantum mechanics
- Conducted numerical approximations and asymptotic studies
- Implemented **numerical simulations** and algorithms in Julia/Python
- Optimized code for efficiency, scalability, and performance in Julia/Python

PhD student at Technical University of Denmark, Lyngby

September 2021 — November 2024

- Created **theoretical models** to describe complex quantum systems and generate insights for experimental validation
- Designed **statistical algorithms** to analyze quantum correlations in emission spectra
- Worked **independently** and in **interdisciplinary** teams

Project Consultant at Tenuta la Favola, Noto

January 2023 — January 2024

Part-time job.

- Planned and **coordinated multiple** projects, establishing clear timelines
- Develop **Python automation** tools to streamline financial data analysis, improving accuracy
- Monitored and evaluated financial performance by analyzing multi-year expense and trends
- Identified high-impact business sectors to guide strategic decision-making and resource allocation

PhD visitor at Technical University of Berlin, Berlin

June 2022 — September 2022

- Collaborated with a multidisciplinary research team on complex projects
- Mentored students in their research projects

Technical Assistant at Search On Media Group S.r.l., Bologna

June 2021

Private Math and Physics tutor at Freelance, Bologna

January 2017 — January 2021

- Mentored
- Explained complex scientific concepts in clear, accessible ways



EDUCATION

Master Degree in Physics, Alma Mater Studiorum, Bologna IT

2018 — 2021

Theoretical Physics, Grade 110/110

Master Two Erasmus Student, Centre de Physique Théorique, Aix Marseille Université, Marseille FR

2019 — 2020

Theoretical Physics

Bachelor Degree in Physics , Alma Mater Studiorum, Bologna IT

2014 — 2017

Physics, Grade 104/110

High School Diploma , Liceo Classico Rudiní, Noto IT

2008 — 2013

Classical Studies, Grade 100/100

High School Diploma, Saint Helena High School, Saint Helena USA

2011 — 2012



REFERENCES

- Kristín Björg Arnardóttir from Molecular Quantum Solutions



PUBLICATIONS

Dynamics and condensation of polaritons in an optical nanocavity coupled to two-dimensional materials

Physical Review B, Vol. 109, 155432,

M. V. Gurrieri, E. V. Denning, K. Seegert, P. T. Kristensen, J. Mørk



CONFERENCES

PEPS, Reykjavik, IS

Polariton Science Conference, Aarhus, DK

OECS — CNR-NANOTEC, Lecce, IT

DPG , Regensburg, DE

Polaron Day , St. Andrews, UK